

# Waste-IQ Project Report

## AI-Powered Waste Recycling Marketplace

Waste-IQ is a recyclable waste pickup marketplace that connects citizens, collectors, and administrators through a centralized platform.

### Objectives

Digitize recyclable waste collection, improve collector efficiency, and provide administrators with analytics and operational visibility.

### Technology Stack

Frontend: React, Vite, Tailwind CSS  
Backend: FastAPI, SQLAlchemy, JWT Authentication  
Database: PostgreSQL  
Deployment: Docker, Render

### Implemented Features

- User Registration & Login
- Role-Based Access Control
- Citizen Pickup Requests
- Collector Accept & Complete Workflow
- Admin Analytics Dashboard
- Docker & PostgreSQL Support

### Future Enhancements

- AI Waste Image Classification
- Geo-Matching of Collectors
- Waste Value Prediction
- WhatsApp Notifications
- Demand Forecasting